Sujal Mallick

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Education

Thapar Institute of Engineering & Technology — Patiala, Punjab
Bachelor of Engineering in Computer Engineering; CGPA: 7.81/10 — *Aug 2023 – May 2027*

ODM Public School, Bhubaneswar | *Senior Secondary (CBSE), Class 12: 91%* — *May 2023*

Experience

Aumbram — Patiala, Punjab
QA & CDN Development Intern — *Jun 2026 – Present*

- Engineered the CDN/media pipeline for a video-learning platform on Azure: presigned multipart uploads (up to 2 GB), FFmpeg HLS transcoding into **5** adaptive-bitrate renditions, and HMAC-signed playback URLs
- Implemented event-driven transcoding workers on Redis queues that KEDA scales from zero (about 5 jobs per replica), with a transactional outbox, idempotent jobs, and 3 backoff retries so no upload is processed twice or lost
- Constructed a **6-state** PostgreSQL-backed state machine with a fail-closed moderation gate, blocking **100%** of unreviewed videos from publishing across upload, processing, and publish stages
- Analyzed 7+ Android/iOS builds of a React Native app and uncovered **152 defects** (44 high-priority), documenting severity and repro steps for developers; **82%** were fixed

SigMap (Open Source) | **GitHub** — Remote
Open Source Contributor — *Oct 2026*

- Shipped 2 merged language extractors (PowerShell, Objective-C/C++) to SigMap, a code-signature mapper that shrinks LLM context for AI coding agents (**645+** GitHub stars, **13K+** monthly npm downloads)
- Developed the PowerShell extractor across **5** constructs (functions, parameter blocks, comment-based help, classes, module manifests); masked method bodies to stop phantom and leaking signatures
- Authored the Objective-C extractor for **3** file types (.m, .mm, .h) covering interfaces, protocols, categories, and multi-part selectors, with content detection and a safe fallback for MATLAB .m files

Projects

BugMind AI | *React, FastAPI, PostgreSQL, LangGraph, Azure* | GitHub | Live Demo — Jun 2026 – Present

- Architected and independently shipped an AI QA platform on Azure: **131** REST endpoints, **411** automated tests, and **50K+** lines of Python/TypeScript across 47 merged PRs with GitHub Actions CI/CD
- Orchestrated **8 LLM agents** that turn plain-text requirements into 6 QA artifacts (modules, checklists, test cases, test plans, E2E scripts, bug reports), with a LangGraph pipeline that loops back to fill coverage gaps
- Secured multi-tenant organizations with a **4-level** role hierarchy (viewer to owner), email invites, and encrypted bring-your-own model keys

Swarm-CUDA: GPU Flocking Simulator | *C++, CUDA, OpenGL, Thrust* | GitHub — Apr 2026

- Parallelized the simulation core in CUDA (**3** steering kernels, GPU agent state, update loop) for a 4-person team's simulator that handles up to **200K** agents at interactive frame rates
- Unblocked Windows builds by fixing the CMake configuration for MSVC; paired the core with the team's spatial-hash neighbor search (Thrust GPU sort) and CUDA–OpenGL rendering

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, SQL, CUDA
Backend & Web: FastAPI, SQLAlchemy, PostgreSQL, Redis, React.js, Node.js, Express.js, Tailwind CSS
Cloud & DevOps: Azure (App Service, Container Apps), AWS (EC2, S3), Docker, KEDA, GitHub Actions
AI/LLM: LangGraph, LangChain, LiteLLM, OpenAI API, Gemini API
Testing: Pytest, Regression & Mobile QA

Leadership

Fine Arts & Photography Society, Thapar University — 2023 – 2025
Core Member: mentored 40+ juniors in recruitment; led design and videography for 6+ events